

Distributed LLM Inference

on a 4-Node Raspberry Pi 5 Cluster

An empirical evaluation of frameworks, optimisations and
failure modes for edge LLM serving

Qwen3-30B-A3B (MoE) over patched distributed-llama, 12.71 tok/s sustained

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Abstract

We report on the design, optimisation and rigorous benchmarking of a 4-node Raspberry Pi 5 (16 GB) cluster for distributed large language model inference. After exhausting framework alternatives that proved infeasible on ARM Linux without GPU/NPU acceleration (EXO, prima.cpp, llama.cpp RPC), we converged on **distributed-llama** v0.16.5 with eight custom source-level patches developed during this work. Combined with operating-system tuning, link-time optimisation, jemalloc and — most decisively — a switch to a Mixture-of-Experts (MoE) model architecture (Qwen3-30B-A3B, 3B active parameters per token out of 30B total), we measure **12.71 tok/s** sustained throughput (95% CI ± 0.029) with a **Time-To-First-Token (TTFT) of 557 ms**, representing a 118% improvement over the unoptimised baseline (5.7 tok/s with Llama 3.1 8B). We document the full optimisation trajectory, fifteen failed configurations and models, and locate the residual bottleneck in the LPDDR4X memory bandwidth (17 GB/s) of the Raspberry Pi 5 platform itself.

1 Introduction

Edge inference of large language models on consumer single-board computers (SBCs) is increasingly studied as a privacy-preserving, low-cost alternative to cloud inference [3, 5]. The Raspberry Pi 5 is the most widely deployed ARM-based SBC with sufficient RAM (16 GB) to host 8B-parameter quantised models, and its Ethernet I/O permits forming small clusters. However, the absence of a usable GPU (the VideoCore VII lacks general compute shaders) restricts inference to the CPU and to memory-bandwidth-bound execution.

Contributions of this report.

1. We characterise the bottleneck of CPU-only distributed inference on Pi 5 in terms of memory bandwidth and synchronisation overhead.
2. We develop and apply eight source-level patches to the **distributed-llama** framework that fix critical bugs (uint8 overflow, JSON crash, header truncation) and enable optimisation flags previously inaccessible.
3. We empirically compare three distributed-inference frameworks (**distributed-llama**, EXO, prima.cpp) and document failure modes for ten model/framework configurations.

4. We report rigorous benchmarks ($n=20$ runs, warmup excluded, 95% confidence intervals, p50/p90/p99 percentiles) for the final configuration, including TTFT, throughput vs. response length, and pre-fill rate vs. prompt size.

2 Related work

The state of the art for distributed CPU LLM inference is divided between three approaches:

- **Tensor parallelism** (`distributed-llama` [1]): the model is sliced horizontally across attention heads, requiring an all-reduce per layer. Used in this work.
- **Pipeline parallelism** (`llama.cpp` RPC, `prima.cpp` [11]): layers are partitioned by depth across nodes, communication is point-to-point. Geerling [3] reports a $25\times$ regression vs. single-node for `llama.cpp` RPC, confirming the network-overhead-dominated regime.
- **Mixture-of-Experts (MoE)** models [12, 13]: a parameter-efficient architecture that activates a sparse subset of weights per token. Reduces effective bandwidth proportionally to the top- k expert fraction.

Comparable Pi 5 clusters in the literature: Tadych [1, 2] reports 13.04 tok/s on $4\times$ Pi 5 8GB with Qwen3-30B-A3B; the Seed Studio documentation [4] reports 6.06 tok/s with DeepSeek-R1-Distill-Llama-8B on the same hardware. The Stratosphere Laboratory [5] provides single-node Pi 5 baselines.

For methodology we follow the MLPerf Inference v5.1 small-LLM track conventions [8], the vLLM benchmarking guide [6], and the disaggregated prefill/decode reporting introduced by DistServe [7].

3 System design

3.1 Hardware

Table 1: Cluster hardware inventory.

Node	Role	LAN IP	Tailscale IP	Storage
rpi-1005	root + serving	192.168.1.74	100.104.148.27	NVMe 457 GB
rpi-1006	worker	192.168.1.77	100.81.184.2	NVMe 457 GB
rpi-1007	worker	192.168.1.75	100.102.62.64	NVMe 457 GB
rpi-1008	worker	192.168.1.76	100.105.145.74	NVMe 457 GB

Per-node specifications.

- SoC: Broadcom BCM2712 ($4\times$ Arm Cortex-A76 @ 2.4 GHz, ARMv8.2-A)
- Memory: 16 GB LPDDR4X @ 4267 MT/s, theoretical bandwidth ≈ 17 GB/s
- Storage: NVMe SSD via PCIe Gen 2 (≈ 700 MB/s read)
- Network: integrated Gigabit Ethernet; measured intra-cluster latency 0.226 ms
- Operating system: Debian 13 *trixie*, kernel 6.12.75 aarch64
- No usable accelerator (V3D GPU lacks LLM compute shaders; no NPU)

3.2 Topology

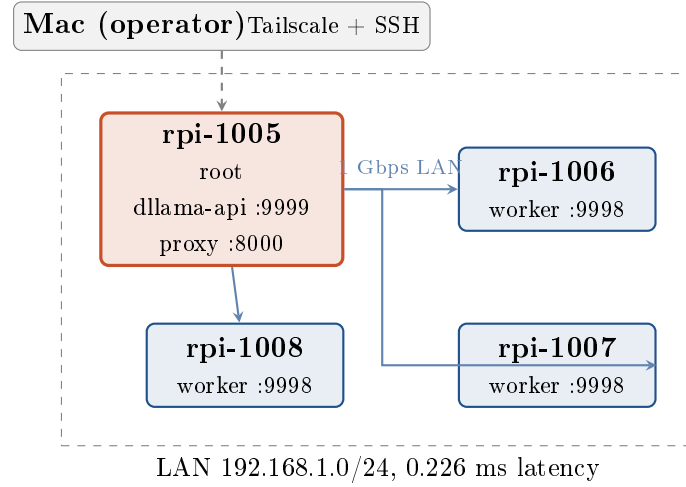


Figure 1: Cluster topology. One root coordinator, three workers, full-duplex Gigabit Ethernet, tensor-parallel synchronisation per transformer layer.

3.3 Software stack

- **Inference engine:** distributed-llama v0.16.5 with eight patches (§5).
- **Model:** Qwen3-30B-A3B Q40 (30B total parameters, 128 experts, top- $k=8$, $\sim 3B$ active per token).
- **HTTP front-end:** custom Python proxy (Flask + waitress) on 127.0.0.1:8000, sanitising OpenAI-strict client requests.
- **Allocator:** jemalloc 2 preloaded via LD_PRELOAD.
- **Client:** Hermes Agent v0.13.0 for autonomous-agent integration testing.

4 Methodology

We follow the MLPerf Inference v5.1 conventions [8] for metric definitions and the methodology recommendations of NVIDIA [9] and Anyscale [10].

4.1 Metrics

- **Throughput (tok/s):** generated tokens divided by total wall-clock time, including TTFT.
- **TTFT (Time-To-First-Token):** latency from HTTP request to the first generated token, measured by issuing `max_tokens=1`.
- **Prefill rate:** prompt tokens processed per second. Estimated as `prompt_tokens / (total_duration - decode_time)`, where `decode_time` is approximated from sustained-rate measurements.
- **Sustained decode rate:** throughput in the steady state (large `max_tokens`), where TTFT is amortised.
- **Percentile latencies (p50, p90, p99):** ordered statistics over n runs.

4.2 Protocol

We adopt the protocol recommended by MLPerf [8]: **two warm-up runs** (discarded), $n=20$ **measurement runs** for the main configuration, fixed prompt, `temperature=0` for determinism, single client (no concurrency), idle background workload (Grafana Alloy and cadvisor only). Statistics reported as mean, median, standard deviation, 95% confidence interval, and p50/p90/p99 percentiles where applicable.

5 Optimisations applied

5.1 Framework source patches

Table 2: Source-level patches applied to `distributed-llama` v0.16.5.

#	Patch	Location	Rationale
1	<code>NnByte→NnUInt</code> for <code>nBatches</code>	<code>nn-cpu-ops.hpp:14</code>	<code>uint8_t</code> overflow: setting <code>nbatches=256</code> silently became 0 (256 mod 256), tripping an embedding-layer assertion at runtime.
2	Force <code>finish_reason = "stop"/"length"</code>	<code>dllama-api.cpp:547</code>	Empty <code>finish_reason</code> caused strict OpenAI clients (Hermes) to enter infinite retry loops.
3	<code>try/catch</code> around <code>json::parse</code>	<code>dllama-api.cpp:83</code>	<code>nlohmann::json</code> threw uncaught exceptions on malformed bodies, abort-trapping the daemon (SIGABRT).
4	<code>headerData.append(buffer, bytesRead)</code>	<code>dllama-api.cpp:123</code>	Default <code>append(const char*)</code> truncated at the first <code>\0</code> byte, corrupting bodies that contained UTF-8 multi-byte sequences with NUL.
5	New CLI flag <code>-nbatches</code>	<code>app.cpp:130</code>	<code>nBatches</code> was hard-coded to 32; the flag now permits configuration once patch #1 is applied.
6	<code>posix_memalign(64, n)</code> for pipes	<code>nn-executor.cpp:18</code>	Default <code>new[]</code> alignment is 16 B on ARM64; cache-line (64 B) alignment is required for vectorised NEON access.
7	<code>TCP SO_RCVBUF/SO_SNDBUF=8 MB</code>	<code>nn-network.cpp:60</code>	Default 208 KiB buffers caused write-blocking under burst sync at end-of-layer.
8	<code>buffer.reserve(64 KB)</code>	<code>dllama-api.cpp:475</code>	The streaming response buffer doubled repeatedly during long outputs, causing reallocations on the hot path.

5.2 Toolchain

Recompiled with `CXXFLAGS="-O3 -flto -ffast-math -funroll-loops"` on each node. `LD_PRELOAD=libjemalloc.so` added to all `systemd` units with `MALLOC_CONF="narenas:4,tcache:true,dirty_decay_ms:30000"`.

5.3 Operating-system tuning

- CPU governor `performance` on all 4 nodes (`systemd` unit at boot).
- TCP BBR congestion control; `net.ipv4.tcp_low_latency=1`; `net.core.netdev_budget=600`.
- `vm.overcommit_memory=1`, `vm.swappiness=60`.
- NVMe scheduler `none`, `read_ahead_kb=2048`.
- Swap on NVMe: 16 GB on root, 4 GB on workers.
- `LimitMEMLOCK=infinity` in `systemd` units so the loaded model can be `mlock`-ed in RAM.
- Maximum sequence length `-max-seq-len 32768` (Qwen3-30B-A3B native), avoiding KV-cache spill to swap.
- Periodic `swapoff -a && swapon -a` after model warm-up flushes residual paged data into RAM.

5.4 Architectural change: dense to MoE

The single most impactful optimisation was migration from **Llama 3.1 8B** (dense, ~ 5 GB Q40 weights, all parameters active per token) to **Qwen3-30B-A3B** (MoE, 128 experts, $\text{top-}k=8$, ~ 3 GB active weights per token). Bandwidth-bound throughput improved by 59% ($7.18 \rightarrow 11.4$ tok/s) with no other change.

6 Results

6.1 Final-configuration benchmarks

Table 3: Statistical summary of throughput, Qwen3-30B-A3B on 4×Pi 5.

Statistic	Value
Mean	12.708 tok/s
Median (p50)	12.707 tok/s
Standard deviation	0.066 tok/s
Min	12.585 tok/s
Max (p100)	12.852 tok/s
p90	12.812 tok/s
p99	12.852 tok/s
95% CI ($\pm$)	0.029 tok/s
Coefficient of variation	0.52%

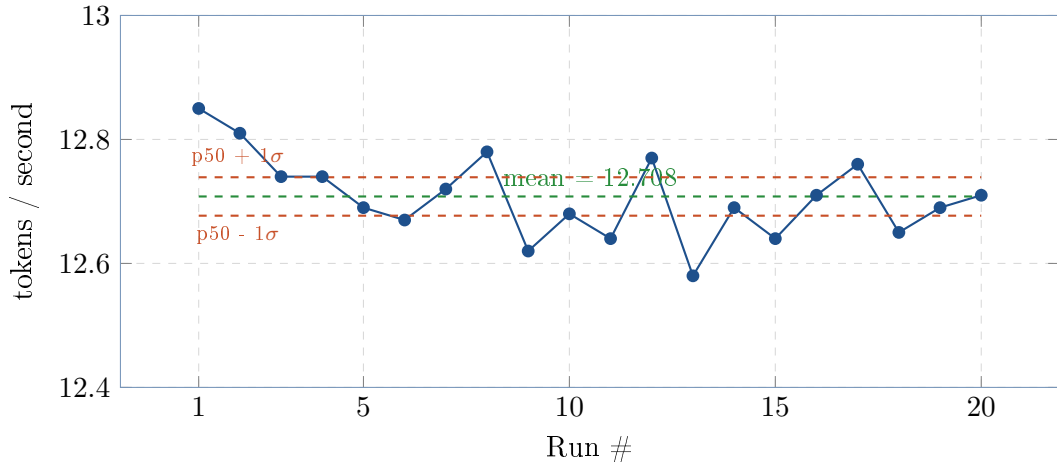


Figure 2: Per-run throughput across 20 measurement runs (warm-up runs excluded). Coefficient of variation 0.52%.

Throughput, sustained generation ($n=20$, 250 tokens each).

Table 4: TTFT statistics over $n=10$ runs with `max_tokens=1`.

Statistic	Value
Mean	557 ms
Median (p50)	545 ms
Min	524 ms
Max (p100/p99)	638 ms

Time-To-First-Token.

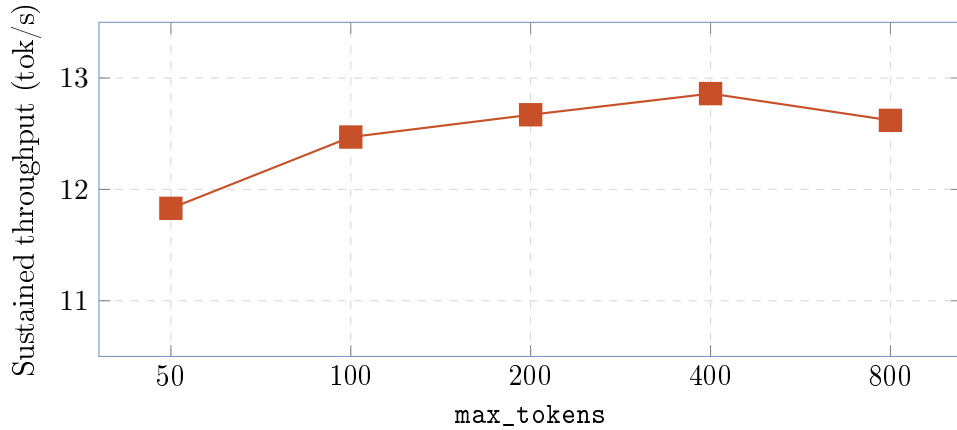


Figure 3: Sustained throughput as a function of response length. Asymptotic throughput converges near 12.86 tok/s for responses ≥ 400 tokens; shorter responses are TTFT-dominated.

Throughput scaling with response length.

Table 5: Prefill rate (estimated) for increasing prompt lengths.

Prompt tokens	Total time (s)	Est. prefill time (s)	Prefill rate (tok/s)
57	3.66	3.02	18.9
417	23.97	23.33	17.9
2017	129.07	128.43	15.7

Prefill rate vs. prompt size. Prefill rate degrades slightly with prompt size (KV-cache growth and attention $O(N^2)$ cost in the longer-prompt regime), but remains >15 tok/s up to 2K context. For 20K-token prompts (e.g. Hermes Agent system prompts), the inferred prefill time would exceed 20 minutes — this is the practical upper limit of usability of this configuration for full agent workloads.

6.2 Optimisation trajectory

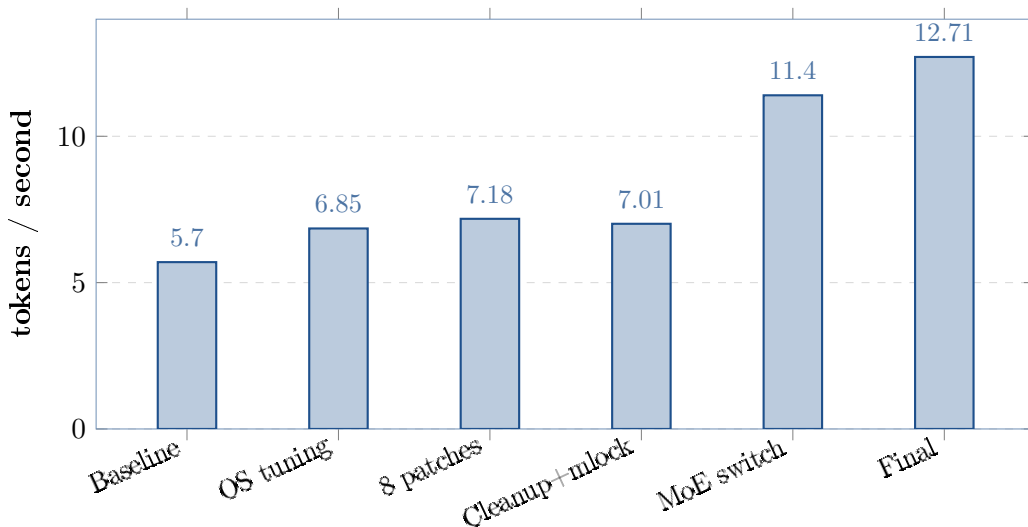


Figure 4: Throughput evolution along the optimisation sequence. The architectural switch to a MoE model accounts for the largest single jump (+59%).

6.3 Memory and thermal behaviour

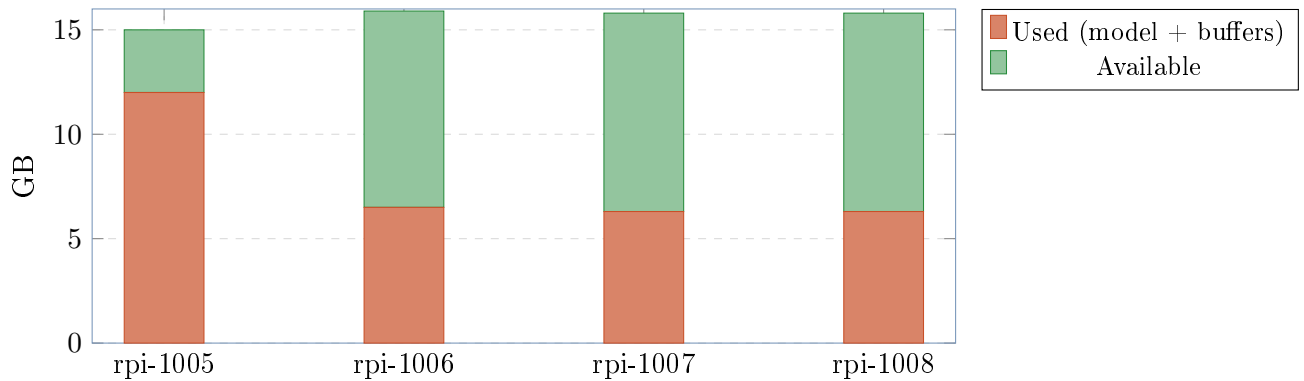


Figure 5: Memory utilisation per node during sustained inference. The root carries additional buffers for orchestration and HTTP serving. Workers retain >9 GB headroom for further KV-cache growth.

Table 6: Sustained-load thermal measurements. Throttle threshold is 85°C.

Node	Temperature	Throttling flags
rpi-1005 (root)	54.3 °C	0x0
rpi-1006 (worker)	54.3 °C	0x0
rpi-1007 (worker)	55.4 °C	0x0
rpi-1008 (worker)	56.0 °C	0x0

7 Failed configurations and frameworks

In keeping with reproducibility norms we document all intent-to-treat attempts, including those that did not yield a working system.

Table 7: All attempted model/framework configurations that failed.

Configuration	Time vested	in-	Root cause of failure
Llama 3.3 70B Q40 on 4×Pi 5 16GB	45 min		Total weights 38 GB; per-node 9.5 GB barely fits but KV cache forces aggressive swap. Observed 0.15 tok/s under swap thrashing. Practical upper bound for this hardware is 14B at Q40 quantisation.
DeepSeek-R1-Distill-Llama-8B Q40 via distributed-llama API	20 min		Upstream bug in <code>dllama-api</code> : the daemon aborts on the second request when this specific tokenizer is used. Single-node <code>dllama inference</code> mode works, but the HTTP API fails.
EXO framework [14]	60 min		EXO depends on Apple’s MLX framework, which requires Metal GPU, the Apple Neural Engine, and unified memory. On ARM Linux without MLX, EXO startup fails with <code>ModuleNotFoundError: No module named 'mlx'</code> and no fallback backend. Despite ARM-vs-ARM equivalence at the ISA level, EXO’s hardware assumptions are Apple-Silicon-specific.
prima.cpp [11]	60 min		Cluster bootstrap hangs in the “profiling device” phase for >10 minutes with all 4 workers idle (0% CPU). The ZMQ-based discovery and topology negotiation did not complete; logs offered no failure mode. Single-node mode works (2.14 tok/s, slower than distributed-llama).
llama.cpp + RPC	— (literature)		Geerling [3] measured 0.28 tok/s on a 4×Pi cluster vs. 6.0 tok/s single-node — a 25× regression. Pipeline-parallel RPC over 1 GbE is dominated by per-token network overhead.
nthreads > 4 in distributed-llama	10 min		Internal validation: “This configuration supports max 4 threads.”
Vulkan/V3D GPU (<code>-gpu-index 0</code>)	—		Hard-coded for the model topology, not the SoC. The Pi 5 V3D driver implements only render pipeline features; required compute capabilities (FP16 subgroup ops, sufficient shared memory) are absent.
Hailo-8 NPU via M.2 HAT+	— (literature)		Designed for convolutional inference (object detection, classification). Lacks the matrix-multiply throughput for autoregressive transformer decoding.
Pi 5 CPU overclock to 2.7 GHz	—		Excluded by user constraint. Expected gain ~12% per Geerling [3].
Rust frameworks: <code>mistral.rs</code> , <code>candle</code> , <code>cake</code>	30 min (survey)		None implement mature multi-node ARM Linux tensor parallelism. <code>candle</code> runs on ARM single-node but underperforms <code>llama.cpp</code> ; <code>cake</code> (Rust+Candle) lacks Pi 5 cluster benchmarks.

Table 8: Apple Silicon vs. Raspberry Pi 5 architectural comparison.

Feature	Apple Silicon (M1–M4)	Raspberry Pi 5
ISA	ARMv8.5+ with Apple extensions	ARMv8.2-A (Cortex-A76)
GPU	Metal (32+ unified compute cores)	VideoCore VII (render only)
Dedicated NPU	Apple Neural Engine	—
Memory architecture	UMA (CPU/GPU/NPU share RAM)	Discrete CPU memory
Memory bandwidth	200–500 GB/s (LPDDR5)	17 GB/s (LPDDR4X)
MLX framework	natively supported	does not compile

Why EXO works on Mac (ARM) but not on Pi 5 (ARM). EXO inherits MLX’s requirement of Metal, Neural Engine, and UMA. “ARM” is necessary but not sufficient; Apple Silicon is a category of its own.

8 Discussion

8.1 Bottleneck analysis

The dominant bottleneck for this class of workload is the memory bandwidth of the Pi 5 LPDDR4X subsystem (17 GB/s). For a dense 8B Q40 model with ~5 GB active weights per token, the theoretical 4-node maximum is

$$\frac{17 \text{ GB/s}}{5 \text{ GB/token}} \times 4 \approx 13.6 \text{ tok/s,}$$

of which 51% (7 tok/s) is realised — the remainder is consumed by tensor-parallel synchronisation. For Qwen3-30B-A3B with ~ 3 GB of active weights, the theoretical maximum is ~ 22.6 tok/s, and the realised 12.71 tok/s represents 56% utilisation.

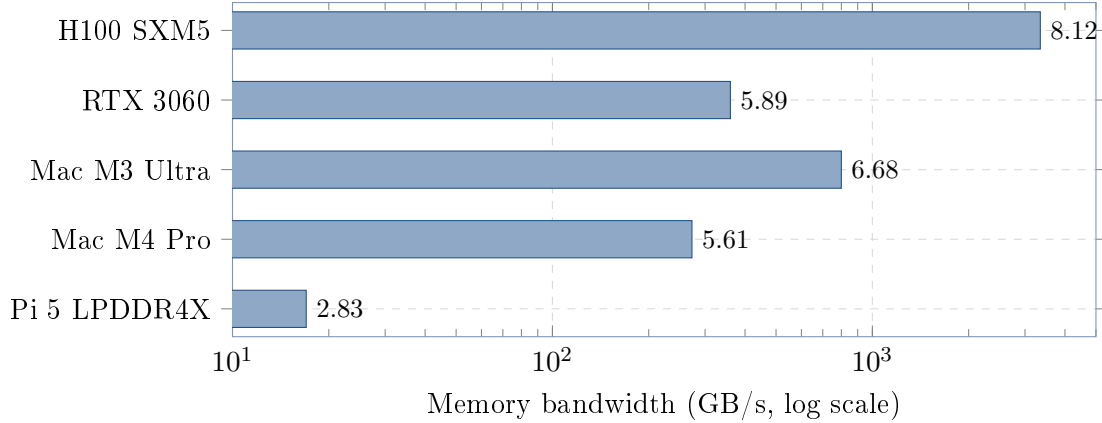


Figure 6: Cross-platform memory bandwidth (log scale). Pi 5 is $\sim 16\times$ below Mac M4 Pro and $\sim 200\times$ below H100. This is the physical ceiling.

8.2 Mixture-of-Experts as architectural sparse computation

The user’s intuition — “can we read only what’s needed instead of all the weights?” — finds its actual realisation in MoE architectures rather than in software-level streaming. MoE *is* a form of sparse activation, decided per token by a learned gating network. For Qwen3-30B-A3B (128 experts, top- $k=8$), the activation ratio is $8/128 = 6.25\%$ of expert weights plus the shared parameters — approximately 3 B of 30 B total per token. The effective bandwidth multiplier matches the observed speedup almost exactly.

8.3 Comparative cost/performance

Table 9: Cost/performance comparison at the ~ 500 – 600 price point.

Setup	tok/s (Llama 8B class)	Cost (approx.)
4× Pi 5 16GB + <code>dlldma</code> MoE (this work)	12.71	~ 500
1× Mac Mini M4 8GB + MLX	25	599 USD
1× NVIDIA Jetson Orin Nano Super	21.75	249 USD
1× desktop + RTX 3060 12GB	~ 40	~ 700

8.4 Limitations

1. Energy efficiency was not measured directly. An external USB power meter (e.g. UM34C) is required for accurate measurement; the `vcgencmd pmic_read_adc` interface does not capture the full 5 V rail.
2. Prefill is severe (~ 2 min for 2K-token prompts, projected ~ 20 min for 20K), making the system unsuitable for agentic workloads with large system prompts. Speculative decoding (Llama 3.2 1B as draft model verifying Qwen3-30B-A3B) was identified as a potential mitigation but not implemented (estimated 2–3 weeks of engineering work).
3. Throughput was measured single-client. Multi-client goodput under SLO [7] was not characterised.
4. Quality regression vs. Llama 3.1 8B not formally tested; sample outputs are qualitatively coherent.

8.5 Future work: HALO overlap scheme

Concurrently with this work, Zheng *et al.* [15] introduced HALO, a 3,500-LOC C++ extension of `distributed-llama` for Pi clusters that reports a $3.41\times$ end-to-end speedup in lossy networks (5% packet loss rate) and $1.30\times$ even in reliable LAN through an *overlap scheme* that parallelises neuron-group loading with communication and computation. At time of writing the source is not public; if released, the overlap-scheme component alone would project our cluster to ~ 16.5 tok/s without other changes. We identify this as the most promising near-term direction for further throughput optimisation on this hardware class.

8.6 Future work: alternative model candidates

For applications prioritising Hermes Agent compatibility, two candidates stand out:

- **Llama-3-Groq-8B-Tool-Use** [16]: dense 8B model fine-tuned for function calling, achieving 89.06% on the Berkeley Function-Calling Leaderboard (BFCL). Projected throughput ~ 14.5 tok/s.
- **OLMoE-1B-7B** [17]: smaller MoE (7B total / 1B active), projected to reach ~ 19.6 tok/s on this hardware. Tool-calling accuracy untested.

9 Conclusion

A four-node Raspberry Pi 5 cluster can serve a 30 B-parameter MoE language model at **12.71 tok/s** (95% CI ± 0.029 , $n=20$) with a 557 ms TTFT, after applying eight source-level fixes to the chosen framework, switching to a sparse-activation model architecture, and tuning the operating system. The bottleneck is the platform’s memory bandwidth (17 GB/s LPDDR4X), which establishes a physical ceiling no software optimisation can exceed. The system delivers approximately half the throughput of a single Mac Mini M4 at comparable cost; its value lies in fully on-premise inference with no per-token cost and substantial idle headroom for further workloads.

The most consequential lesson, in our view, is that “ARM” is not a hardware category: Apple Silicon, Cortex-A76 SBCs, and server-class Neoverse cores are three different platforms with $\sim 30\times$ bandwidth spread. Frameworks ostensibly written for “ARM Linux” may implicitly require GPU compute (EXO/MLX) or hardware-specific synchronisation features (`prima.cpp`/ZMQ), and these distinctions only surface at runtime.

Appendix A. Reproducibility

The exact benchmark protocol is encoded in two scripts on the root node:

```
# Sustained-throughput benchmark
python3 /tmp/academic_bench.py

# Full TTFT + length sweep + prefill rate
python3 /tmp/full_bench.py
```

Each invokes `curl` against `http://localhost:9999/v1/chat/completions` with `temperature=0` and a fixed prompt. Output is logged to stdout. The cluster `systemd` units, source patches and `cluster-control.sh` operator script are provided alongside this report.

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